HARPREET SINGH

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer and Senior Software Engineer with 8+ years of experience specializing in artificial intelligence, Python backend architecture, and production-grade LLM implementations. Over the last 3 years, built and scaled enterprise Generative AI systems using Retrieval-Augmented Generation (RAG), pgVector, semantic search embeddings, and dynamic chunking strategies, alongside computer vision pipelines. Backed by deep expertise across Angular (v2-v20), Vue.js, and FastAPI, delivering scalable full-stack enterprise platforms, microservices, and high-performance user interfaces.

TECHNICAL SKILLS

AI & Machine Learning: Retrieval-Augmented Generation (RAG), LLMs (OpenAI, Anthropic), Vector Databases (pgVector), Embeddings, Semantic Search, Chunking Strategies, Prompt Engineering, Computer Vision (YOLOv8, InsightFace)

AI-Assisted Development: Claude CLI (advanced)

Languages: Python (Asyncio, FastAPI), TypeScript, JavaScript, SQL, Core Java, HTML5, CSS3

Backend & Data: FastAPI, Node.js, PostgreSQL, Oracle DB, MinIO Storage, EDI (cXML, X12 997, SFTP)

Frontend Frameworks: Angular (v2-v20), Progressive Web Apps (PWA), Vue.js, RxJS, NgRx, Ionic, Tailwind CSS, Bootstrap

DevOps & Tools: Docker, Git, Linux Systemd Daemons, Android Studio, Jira, Postman, Visual studio

WORK EXPERIENCE

Signity Software Solutions Pvt. Ltd. | Punjab, India — Jan 2023 - Present

AI Engineer & Senior Software Engineer

- **Wizdom AI (GenAI / RAG Platform):** Architected and deployed an enterprise Generative AI platform orchestrating LLMs with PostgreSQL pgVector; designed custom embedding pipelines and recursive chunking strategies to enable fast, accurate semantic search retrieval across a large enterprise document corpus.
- **Vision-AI (CCTV Surveillance Backend):** Built a real-time computer vision surveillance backend using Python, FastAPI, YOLOv8 face detection, and InsightFace embeddings, processing multi-channel RTSP video from 20 concurrent camera feeds with MinIO storage and streaming telemetry to a Vue.js analytics dashboard.
- **Actus / Intramall (Enterprise B2B Modernization):** Modernized a legacy Java/JavaScript enterprise procurement portal into an automated Python and Oracle DB architecture; developed secure cXML PunchOut integrations, X12 997 EDI handlers via SFTP, and dual-run dynamic SQL schedulers.
- **FenFyre (Multi-Tenant CRM):** Developed a multi-tenant enterprise CRM using Python/FastAPI on the backend and Angular 20 on the frontend, integrating Crypto.js encryption for role-based security layers.
- **SMAKI (E-Commerce Platform):** Spearheaded frontend architecture leading a 4-member UI team within a 10-member cross-functional team; traveled to Germany for an 11-day client-site technical integration.
- **Youth Action Platform (Global Community Portal):** Architected the live platform in Angular, supporting 5 languages (i18n) across a 15-member team.
- **SVB Banking Portal:** Designed frontend architecture for a German banking administration platform with a 6-member team in Angular, implementing Crypto.js client-side data encryption.

IBM | Bengaluru, Karnataka, India — Aug 2021 - Jan 2023

Software Developer

- **COMPASS & Firstnet (AT&T):** Executed major enterprise application codebase upgrades from legacy Angular 6/7 to Angular 12; built generic popups, reusable data tables, and integrated Google Maps APIs while eliminating legacy regression bugs.

- **Vitronics (Government Analytics):** Developed an administrative portal in Angular 12 and MongoDB integrating telemetry from mobile plate recognition and edge verification systems.

Carvewing Solutions | Bangalore, India Mar 2019 - Aug 2021

Software Engineer

- **Quest (France Maritime Cargo):** Programmed interactive cargo scheduling views using D3.js and Angular 10, delivering reusable UI modules and generic data grids.
- **Talkative Parents & Dk-Insights (Bayer):** Constructed Progressive Web Applications (PWA) with Angular, Service Workers, and IndexedDB for offline data caching; integrated Azure MSAL SSO authentication and AWS S3 data pipelines.
- **AES & Agrica:** Launched comprehensive supply chain ERP platforms in Angular and REST APIs; implemented drag-and-drop workflows (Dragula) and multilingual support (i18n).

DSB Edu Tech | Ludhiana, Punjab, India Jul 2018 - Dec 2018

Android Developer

- **Empower Youth:** Engineered a native Android job portal application using Java, XML layouts, Material Design components, and Room Database for offline local caching.

AWARDS & CERTIFICATIONS

- **Secret Superstar Award (2026)** - Signity Software Solutions - Recognized for exceptional technical contribution and ownership across AI and enterprise development initiatives.
- **2nd Runner-Up, AI Hackathon (2026)** - Signity Software Solutions - Awarded 3rd place among organizational teams for building an innovative AI solution prototype.
- **Employee of the Month, August 2020** - Carvewing Solutions - Honored for outstanding performance, team leadership, and executing critical codebase migrations.
- **Angular - The Complete Guide (2022 Edition)** - Udemy - Comprehensive certification covering advanced Angular architectures and reactive programming.

EDUCATION

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

Punjab Technical University (PTU) | Graduated 2018